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# Phase-field modeling and Experimental investigation for rapid solidification in wire and arc additive manufacturing

Fuchen Wang [a], Weipeng Chen [a], Dong Wang [a], Hua Hou [a,c], Yuhong Zhao [a,b,d,*]

[a] School of Materials Science and Engineering, Collaborative Innovation Center of Ministry of Education and Shanxi Province for High-performance Al/Mg Alloy Materials, North University of China, Taiyuan, 030051, China
[b] Beijing Advanced Innovation Center for Materials Genome Engineering, University of Science and Technology Beijing, Beijing, 100083, China
[c] School of Materials Science and Engineering, Taiyuan University of Science and Technology, Taiyuan, 030024, China
[d] Institute of Materials Intelligent Technology, Liaoning Academy of Materials, Shenyang, 110004, China

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ABSTRACT

The rapid solidification speed and significant temperature gradient observed in the wire and arc additive manufacturing (WAAM) process result in a deviation of the solid-liquid interface from the equilibrium state assumed by the conventional directional solidification phase field (PF) model. This, in turn, limits the ability of the traditional directional solidification PF model to accurately depict the microstructure evolution in additive manufacturing, where non-equilibrium effects dominate. In this work, we employed a 3D finite element (FE) model to simulate the temperature distribution during the wire and arc additive manufacturing (WAAM) process. Additionally, a modified PF model that applies to additive manufacturing is also used to simulate the dendrite morphology and solute segregation behavior in the WAAM process at different scanning speeds. Experimental validation of the simulation results was also conducted. We observed a significant reduction in dendrite growth velocity and solute segregation with increasing scanning speed, attributable to the combined effects of the temperature gradient $G(t)$ and solidification speed $V_p$. Furthermore, higher scanning speeds were found to increase the primary dendrite arm spacing (PDAS) and average dendrite width. Importantly, our results demonstrated a quantitative agreement between the simulation results using the modified directional solidification PF model and the experimental observations. This approach provides a foundation for selecting and predicting processing parameters and microstructural characteristics under additive manufacturing conditions, facilitating a comprehensive analysis of dendrite growth and solute segregation at different scanning speeds in the WAAM process.

## 1. Introduction

WAAM has unique advantages compared to other additive manufacturing (AM) processes, such as selective laser melting (SLM), fused deposition molding (FDM), and electron beam melting (EBM), for manufacturing large components due to its high deposition rate, low production cost, and consistent product quality. However, the WAAM process is usually accompanied by large melt pool diameters (5–10 mm) and layer heights (1–3 mm), resulting in significant residual stresses [1] and microstructural heterogeneity [2] such as pores, cracks, and large epitaxial grains [3]. To optimize the WAAM process, a quantitative relationship between process parameters-microstructure, that is., macroscopic-microscopic, is required [4]. However, due to the complexity of WAAM process parameters (such as processing power, scanning speed, scanning strategy, etc.) and the sensitivity of microstructure to processing parameters, it is expensive to collect experimental parameters through trial-and-error experiments [5]. Therefore, with the help of multiscale computational materials science [6–9], developing experimentally verified numerical models is a feasible means to assist WAAM in producing products with desired performance.

Based on the existing solidification theory, among the commonly used microstructure simulation methods, the PF method does not need to track the complex solid-liquid interface and can couple PF equations such as flow, solute, and temperature fields [10]. It makes the microstructure simulation more aligned with the engineering reality and has been widely used in microstructure simulation [11–13]. Using the PF

* Corresponding author. School of Materials Science and Engineering, Collaborative Innovation Center of Ministry of Education and Shanxi Province for High-performance Al/Mg Alloy Materials, North University of China, Taiyuan, 030051, China.
*E-mail address:* zhaoyuhong@nuc.edu.cn (Y. Zhao).

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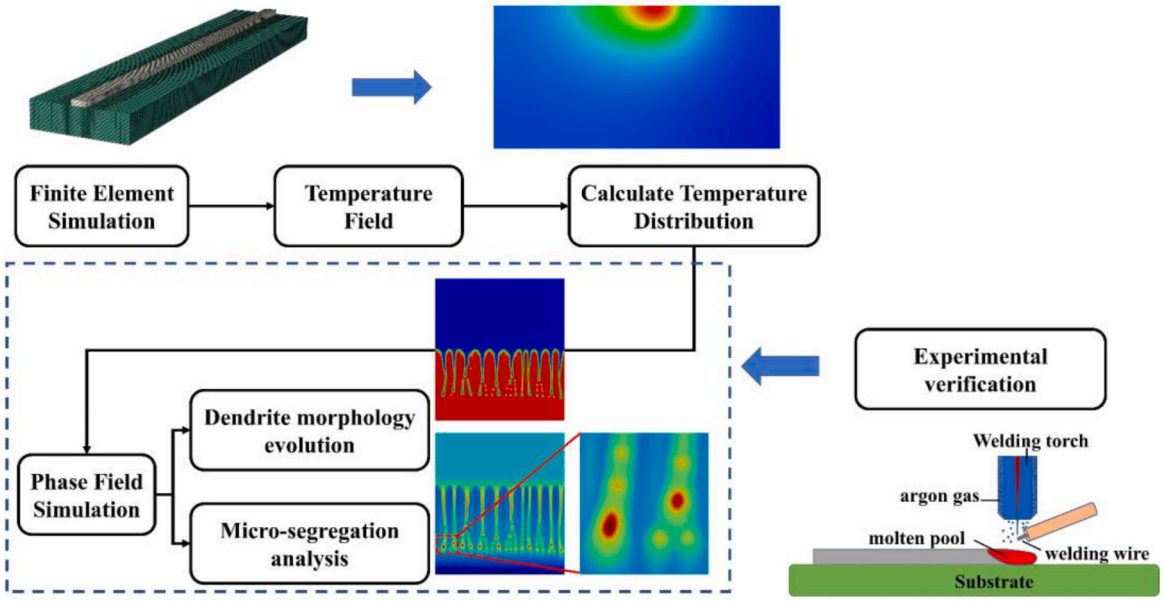



**Fig. 1.** Schematic diagram of phase field simulation combining finite element model and experimental verification.

method, Zheng et al. [14] investigated the dendrite growth during gas tungsten arc welding (GTAW) of 2Al4 aluminum alloy. They established a computational model that better fits the welding process's temperature gradient and solidification speed to make the simulation process more consistent with engineering reality. They successfully predicted the dendrite spacing and the growth direction of the side branches for a given welding parameter. Keller et al. [15] simulated the microstructure evolution of nickel-based superalloys during laser powder bed melting (LPBF) based on FE, PF, and CALPHAD methods. They simulated the solute segregation and dendrite spacing during rapid solidification and achieved good agreement with experimental observations. Geng et al. [16] used a macroscopic FE model to calculate the temperature gradient $G(t)$ for different process parameters. They simulated the dendrite growth process of aluminum alloys under different process parameters by coupling the PF equation to achieve a quantitative link between macroscopy and microscopy. By combining PF and experiments, Liu et al. [17] investigated the epitaxial growth behavior and stray grain formation during laser remelting of nickel-based superalloys. They analyzed the direction of epitaxial growth of matrix grains and the mechanism of stray grain formation. Azizi et al. [18] investigated the effect of build direction on the microstructure during LPBF of Al–Si alloys. They performed PF simulations of grain nucleation and morphological transitions under different build directions by coupling FE thermal models. Bhagat et al. [19] developed a PF model including FE to

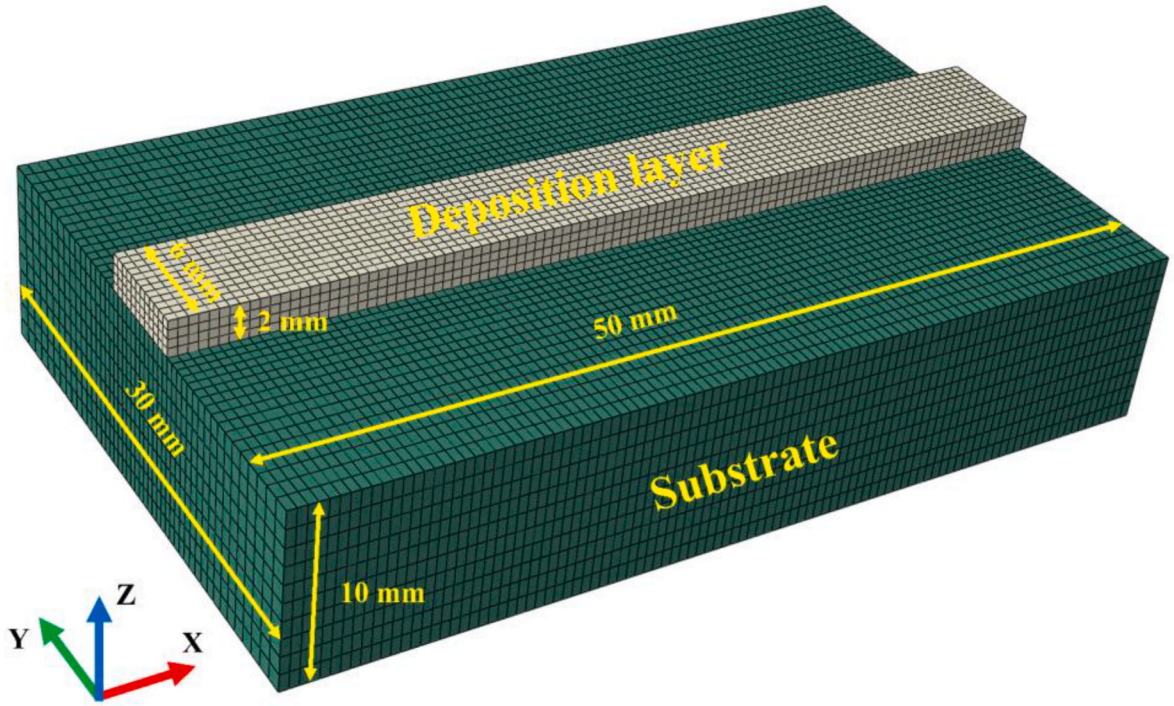



**Fig. 2.** Schematic diagram of 3D finite element model.

simulate the evolution of the melt during metal AM and used this numerical model to predict the solidification of various dendrites. It is worth noting that in the PF model used in the above work, to counteract the artificially introduced "interface thickness" effect in PF simulation, the "antitrapping current" proposed by Karma in the thin interface analysis PF model [20] was introduced. Nevertheless, the Karma model considers that the solid-liquid interface is in local equilibrium. The solute atom diffusion at the solid-liquid interface coincides with the equilibrium phase diagram. When the solid-liquid interface moves at a lower speed, the solute diffusion is dominated by equilibrium effects. Therefore, the Karma model is only applicable for low-speed directional solidification. However, the temperature gradient in the AM process is usually in the range of $10^4 - 10^7$ K/m [21], and the cooling rate is high, with nonequilibrium effects dominating the solidification process [22]. Therefore, the use of the conventional quantitative PF model in the above work to predict the microstructure evolution during rapid solidification under additive manufacturing conditions would cause the solute to be more trapped by the solid-liquid interface during diffusion, which would deviate from the actual results and result in inaccurate simulation results.

In this work, we simulate the dendrite morphology and solute diffusion phenomena during WAAM using a modified PF model [23]. The influence of artificially defined interface thickness on the accuracy of the PF simulation is eliminated. The improved antitrapping current term avoids the influence of the solid-liquid interface on solute diffusion in the PF simulation of rapid solidification. Meanwhile, compared with the experimental data of arc additive manufacturing of ER4043 alloy, it is proved that the modified phase-field model can accurately predict the solute diffusion phenomenon during the arc additive manufacturing process at a higher cooling rate. It makes the PF simulation more consistent with the actual results and provides a feasible strategy for exploring the optimal manufacturing process of WAAM.

## 2. Numerical models and experimental procedures

In this work, we investigated the dendrite growth and solute segregation behavior during WAAM using the PF model established by Kavousi et al. [23] applicable to AM. Fig. 1 shows the process design framework of this work. Due to the high cooling rate in the WAAM process, it is challenging to measure the temperature gradient directly. Therefore, a multiscale model was established in this work by combining a 3D FE model and PF simulation. The temperature gradient in the WAAM process was calculated by a 3D FE model in the macroscopic range, and it was substituted into the PF model as a known parameter to simulate the dendrite growth and solute segregation under different scanning speeds. Finally, the accuracy of simulation results was verified by experiments.

### 2.1. Finite element model

In this work, the temperature gradients at different scanning speeds during WAAM were calculated by a 3D FE model. The 3D FE model, established based on experimental results, is shown in Fig. 2.

The controlling equation for heat transfer from the arc during WAAM can be expressed as,

$$\lambda_c\frac{\partial^2 T}{\partial x^2}+\lambda_c\frac{\partial^2 T}{\partial y^2}+\lambda_c\frac{\partial^2 T}{\partial z^2}+Q=\rho c_p\frac{\partial T}{\partial t}, \tag{1}$$

where $\lambda_c$ is the thermal conductivity, $Q$ is the effective heat source power, $\rho$ is the density, and $c_p$ is the specific heat.

The heat loss caused by convection and radiation between the parts and the external environment is considered in the FE simulation. Newton law describes convective heat transfer as shown in Equation (2). Stefan-Boltzmann law defines radiative heat transfer as shown in Equation (3),

**Table 1**
Parameters used in the finite element simulation [24].

| Simulation parameters | Value | Unit |
|---|---|---|
| Density, $\rho$ | 2700 | kg/m$^3$ |
| Specific heat, $c_p$ | 908 | J/(kg K) |
| Convection heat transfer Coefficient, $h$ | 20 | W/(m$^3$K) |
| Radiant heat transfer Coefficient, $\varepsilon$ | 0.8 | – |
| Ambient temperature, $T_0$ | 20 | $^\circ C$ |
| Stefan-Boltzmann constant, $\sigma$ | $5.67\times10^{-8}$ | W/(m$^2$K$^4$) |

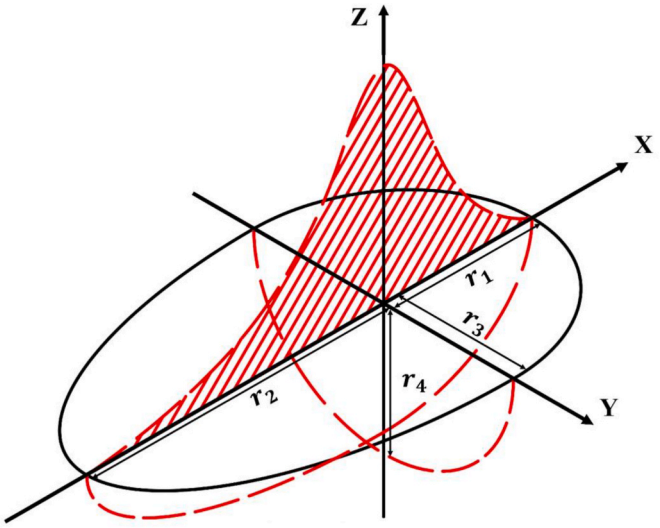



**Fig. 3.** Schematic diagram of double ellipsoidal heat source model.

$$Q_c=-h(T-T_0), \tag{2}$$

$$Q_r=-\varepsilon_r\sigma\left(T^4-T_0^4\right), \tag{3}$$

where $Q_c$ is the convective heat loss, $h$ is the convective heat transfer coefficient, $T$ is the system temperature, $T_0$ is the ambient temperature, $Q_r$ is the radiative heat loss, $\varepsilon$ is the radiative heat transfer coefficient, and $\sigma$ is the Stefan-Boltzman constant. Some of the parameters used in FE simulation are shown in Table 1.

The double ellipsoidal heat source model proposed by Goldak et al. [25] describes the temperature distribution in the WAAM process. The control equation of the double ellipsoidal heat source can be expressed as,

$$q_1(x,y,z)=\frac{6\sqrt{3}Qf_f}{r_1r_3r_4\pi\sqrt{\pi}}exp\left(-\frac{3x^2}{{r_1}^2}-\frac{3y^2}{{r_3}^2}-\frac{3z^2}{{r_4}^2}\right), \tag{4}$$

$$q_2(x,y,z)=\frac{6\sqrt{3}Qf_b}{r_2r_3r_4\pi\sqrt{\pi}}exp\left(-\frac{3x^2}{{r_2}^2}-\frac{3y^2}{{r_3}^2}-\frac{3z^2}{{r_4}^2}\right), \tag{5}$$

where $q_1(x,y,z)$ is the energy equation of the front ellipsoid, $q_2(x,y,z)$ is the energy equation of the back ellipsoid, $Q$ effective heat source power, $f_f, f_b$ are the energy fractions of the front and back ellipsoids, $f_f+f_b=2$, usually $f_f=0.6, f_b=1.4$, $r_1, r_2, r_3, r_4$ are the geometric parameters of the double ellipsoidal heat source. The schematic diagram of the double ellipsoidal heat source model is shown in Fig. 3.

### 2.2. Phase field model

In this work, a modified quantitative PF model [20,23] was used to simulate the dendrite growth and solute segregation behavior in the WAAM process at different scanning speeds. Since the main components of ER4043 wire are Al and Si, the material is assumed to be an Al–Si

**Table 2**
Parameters used in the phase field simulation [26].

| Simulation parameters | Value | Unit |
|---|---|---|
| Equilibrium partition coefficient, $k_e$ | 0.11 | – |
| Liquidus temperature, $T_L$ | 629 | $^\circ C$ |
| Solidus temperature, $T_S$ | 577 | $^\circ C$ |
| Liquidus slope, $m$ | −6.6 | K/mol% |
| Liquid diffusion coefficient, $D_l$ | $3.0 \times 10^{-9}$ | $m^2 s^{-1}$ |
| Solid diffusion coefficient, $D_s$ | $1 \times 10^{-12}$ | $m^2 s^{-1}$ |
| Anisotropy, $\varepsilon_4$ | 0.03 | – |
| Initial concentration, $c_0$ | 5.0 | wt.% |

binary alloy in PF simulation. The parameters used in the PF simulations of Al-5wt.%Si alloy are shown in Table 2. The quantitative PF model introduces a continuous variable $\varphi$ that varies continuously within a finite interface width $W$, with $\varphi = -1$ in the liquid and $\varphi = 1$ in the solid. The temperature distribution of the microstructure during solidification is approximated using the frozen temperature as,

$$T = T_0 + G(t)\left(z - \int_0^t V_p(t^{'})dt^{'}\right), \tag{6}$$

where $G(t)$ is the temperature gradient, $z$ is the direction of dendrite growth.

For the calculation of the temperature gradient, a method for calculating the temperature gradient under nonlinear solidification conditions proposed by Zheng et al. [14] can be expressed as,

$$G(t) = \frac{T_p - T_l}{\sqrt{V^2 t^2 + H^2 \frac{1 - V^2 t^2}{r_2^2}}}, \tag{7}$$

where $T_p$ is the maximum melt pool temperature, $T_l$ is the liquidus temperature of the alloy, $V$ is the scanning speed and $H$ is the height of the melt pool.

The solidification speed $V_p$ is given by,

$$V_p = V \cos\alpha, \tag{8}$$

$\alpha$ is the angle between the interface normal and the scanning speed $V$, as shown in Fig. 4. The PF equation can be expressed as [20],

$$\left(\frac{1 + k_e - (1 - k_e)h(\varphi)}{2}\right)\frac{\partial U}{\partial t} = \vec{\nabla} \bullet \left(D_l q(\varphi)\vec{\nabla}U - \vec{j}_{at}\right) + [1 + (1 - k_e)U]\frac{1}{2}\frac{\partial \varphi}{\partial t}, \tag{10}$$

with

$$U = \frac{1}{1 - k_e}\left[\frac{c/c_l^0}{(1 - \varphi)/2 + k_e(1 + \varphi)/2} - 1\right], \tag{11}$$

$$\tau = \tau_0 a_s(\vec{n})^2\left[1 - \frac{(1 - k_e)(z - V_p t)}{l_T}\right], \tag{12}$$

where $U$ is the dimensionless supersaturation, $c_l^0 = c_\infty / k_e$ is the concentration of the liquid phase when the temperature is $T_0$, $c_\infty$ is the nominal solute concentration, $k_e$ is the equilibrium partition coefficient, $l_T = |m|(1 - k_e)c_l^0 / G(t)$ is the thermal length, $m$ is the liquidus slope, $\tau$ is the relaxation time, and the interpolation function $h(\varphi) = \varphi$.

In the PF simulation, the time and space scales are $\tau_0$ and $d_0$, respectively, with $\tau_0 = a_2 \lambda W^2 / D_l$ and $d_0 = a_1 W / \lambda$, $a_1 = 0.8839$, $a_2 = 0.6267$, space step size $\Delta x = 0.8W = 2.4 \times 10^{-7} m$, the size of the simulation domain is $500\Delta x \times 500\Delta x$, time step size $\Delta t = 0.01\tau_0 = 1.5 \times 10^{-6} s$, $\lambda$ is the coupling constant, $a_s(\vec{n}) = 1 + \varepsilon_4 \cos[4(\theta - \alpha_0)]$ is the interface energy anisotropy [27,28], $\varepsilon_4$ is the strength of the surface tension anisotropy, $\alpha_0$ is the angle of misorientation of the crystal to the coordinate axis, $\theta$ is the angle between the interface normal and the coordinate axis. The interpolation function [29,30] is given by,

$$q(\varphi) = \frac{1 - \varphi + k_e \frac{D_s}{D_l}(1 + \varphi)}{1 + k_e - (1 - k_e)\varphi}, \tag{13}$$

Since the solid-liquid interface width $W$ used in the simulation is larger than the actual interface width, an antitrapping current $\vec{j}_{at}$ is introduced to counteract the retention phenomenon generated by interface thickness on the solute atomic diffusion,

$$\vec{j}_{at} = -a(\varphi)[1 + (1 - k_e)U]\frac{\partial \varphi}{\partial t}\frac{\vec{\nabla}\varphi}{|\vec{\nabla}\varphi|}, \tag{14}$$

$$\tau\frac{\partial \varphi}{\partial t} = \vec{\nabla} \bullet \left[a_s(\vec{n})^2 \vec{\nabla}\varphi\right] + \partial_x\left(\left|\vec{\nabla}\varphi\right|^2 a_s(\vec{n})\frac{\partial a_s(\vec{n})}{\partial(\partial_x \varphi)}\right) + \partial_y\left(\left|\vec{\nabla}\varphi\right|^2 a_s(\vec{n})\frac{\partial a_s(\vec{n})}{\partial(\partial_y \varphi)}\right) + \varphi - \varphi^3 - \lambda(1 - \varphi^2)^2\left(U + \frac{\left(z - \int_0^t V_p(t^{'})dt^{'}\right)}{l_T}\right), \tag{9}$$

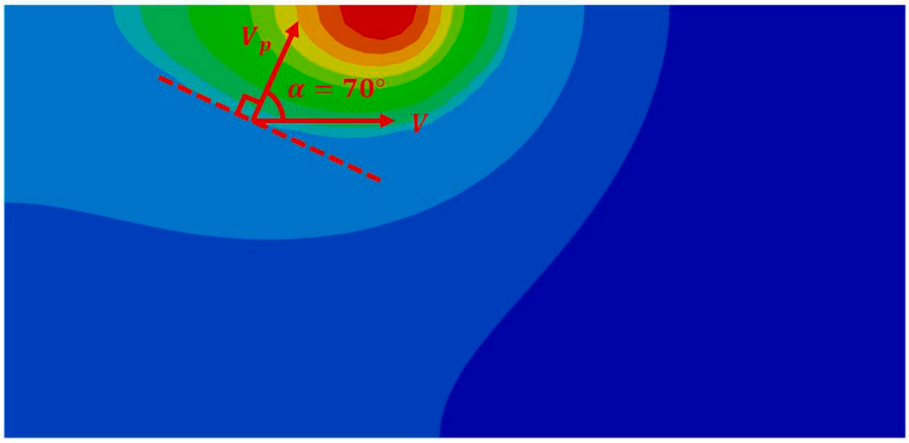



**Fig. 4.** Temperature distribution of the melt pool obtained from the finite element model.

**Table 3**
Chemical compositions of welding wire ER4043 and substrate AA6061 (wt.%).

| | Si | Fe | Cu | Mn | Mg | Zn | Cr | Ti | Al |
|---|---|---|---|---|---|---|---|---|---|
| ER4043 | 5.0 | 0.80 | 0.30 | 0.05 | 0.05 | 0.10 | – | 0.20 | Bal. |
| AA6061 | 0.50 | 0.70 | 0.20 | 0.15 | 1.0 | 0.25 | 0.10 | 0.15 | Bal. |

$$a(\varphi)=\frac{1}{2\sqrt{2}}\left(1-k_e\frac{D_s}{D_l}\right), \tag{15}$$

However, the "antitrapping current" in the Karma model [20] is based on a steady-state case of the equilibrium phase diagram. If the solidification speed is too fast, the diffusion of solute atoms at solid-liquid interfaces is in a non-equilibrium state, which may lead to the formation of supersaturated or metastable phases [31]. Therefore, to satisfy the PF simulation of the WAAM process dominated by non-equilibrium effects, Kavousi and Zaeem [23] modified the anti-trapping current.

The non-equilibrium partition coefficient $k$ in the continuous growth (CG) model [32] is used to replace the equilibrium partition coefficient,

$$k=\frac{\left(k_e+\frac{V_p}{V_D}\right)}{\left(1+\frac{V_p}{V_D}\right)}, \tag{16}$$

where $V_D$ is the solute diffusion velocity [33], and the antitrapping current proposed by Kavousi and Zaeem [23] can be expressed as,

$$a_t(\varphi)=c_1\left(1-\frac{c_2}{ln(1+\varepsilon)}\left(1-\varphi^2\right)\right), \tag{17}$$

$$c_1=\frac{1}{2\sqrt{2}}, \tag{18}$$

$$c_2=\frac{1}{4(1-k_e)}\frac{V_S}{V_D}ln\left(\frac{k_e+1}{2k_e}\right), \tag{19}$$

where $V_S$ is the ratio between the liquid diffusion coefficient $D_l$ and the capillary length $d_0$. A detailed theoretical description can be found in the paper by Kavousi and Zaeem [23]. This study is based on EasyPhase software, which is a comprehensive PF theory software package developed by our research group.

### 2.3. Experimental methods

The samples used in the experiments were fabricated by a dual-inverter AC/DC pulsed TIG arc welder (AVP-500) under the protection of pure argon. The flow rate of argon was 15 L/min, the welding wire is commercial ER4043 wire with a diameter of 1.2 mm, and the substrate is AA6061-T6 alloy. The chemical compositions of the wire and substrate are shown in Table 3 [34]. During the experiments, the angle between the welding gun and the substrate was 80°, the angle between the wire and the substrate was 20°, and the wire feeding speed was 2 m/min. The current during WAAM was controlled at 170 A to obtain a stable morphology, and the scanning speeds were 3 mm/s, 4 mm/s, and 5 mm/s, respectively. At the end of the experiment, samples were taken from the middle part of the deposited layer of WAAM samples, which were polished and then etched with Keller's etchant (HF: HCl: HNO3: distilled water = 1:1.5:2.5:95) for 15 s. The microstructures of the specimens are observed by metallographic microscopy. The microstructure of the samples is observed by SEM at different scanning speeds, and the elemental distribution of the samples is measured by energy dispersive spectroscopy (EDS).

## 3. Results and analysis

### 3.1. Finite element thermal analysis with different scanning speeds

The arc deposition process involves complex processes such as mass and heat transfer, solid-liquid phase transition, etc. A 3D FE model was used to simulate the temperature distribution during WAAM and determine the melt pool's geometry. In the work of Zheng et al. [14], the solidification speed and temperature gradient were closely related to the

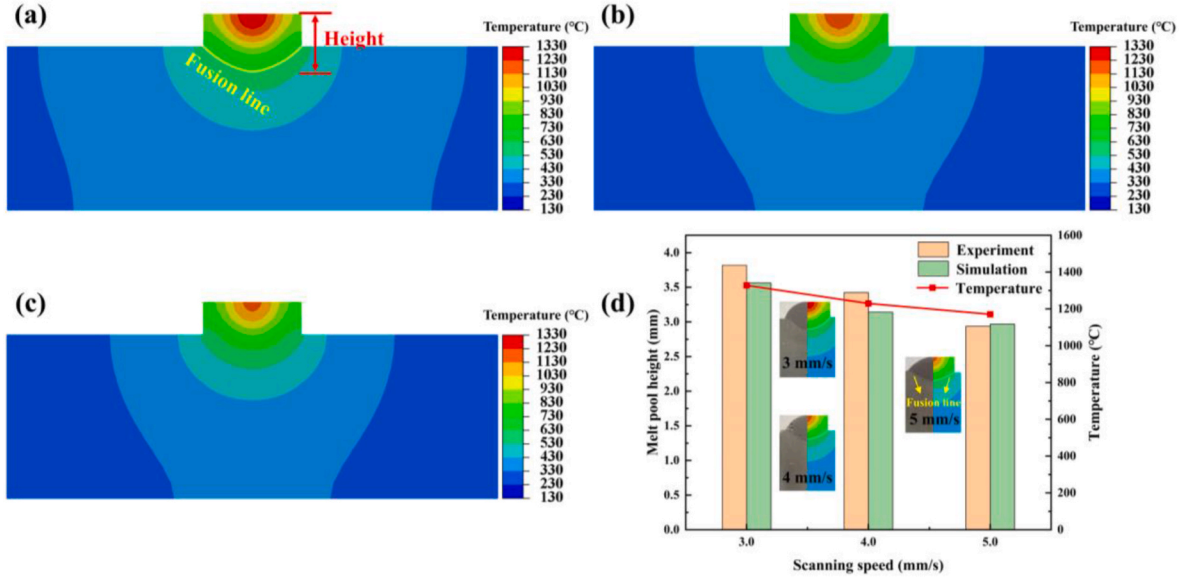



**Fig. 5.** Temperature distribution of the melt pool at scanning speeds of (a) 3 mm/s, (b) 4 mm/s, and (c) 5 mm/s, respectively, (d) Maximum melt pool temperature in (a)–(c) and comparison of melt pool height between finite element simulation and experimental results.

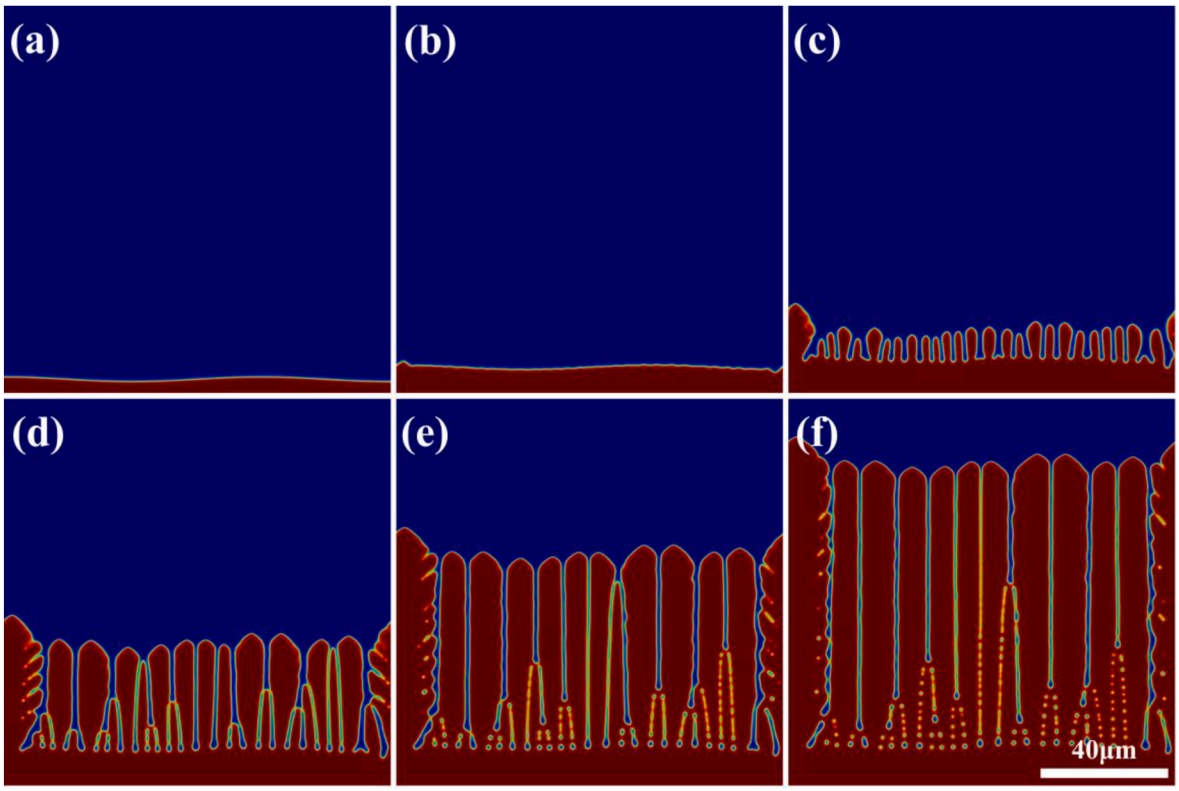



**Fig. 6.** Dendrite growth morphology at (a) $t = 0\Delta t$, (b) $t = 10,000\Delta t$, (c) $t = 15,000\Delta t$, (d) $t = 20,000\Delta t$, (e) $t = 25,000\Delta t$, (f) $t = 30,000\Delta t$ with scanning speed 4 mm/s.

geometry of the melt pool. Since the scanning speed significantly affects the melt pool shape and post-solidification microstructure and is easily adjustable in experiments, the temperature distributions at different scanning speeds were simulated separately. Fig. 5 (a)-(c) shows the schematic diagram of the cross-sectional temperature field along the Y-direction of Fig. 2 at different scanning speeds when the current is 170 A. It can be observed that the temperature of the melt zone decreases with the increase of the scanning speed, mainly because the effective heat input decreases with the increase of the scanning speed. Fig. 5 (d) shows the changes in the melt pool height (as shown in Fig. 5 (a)). The

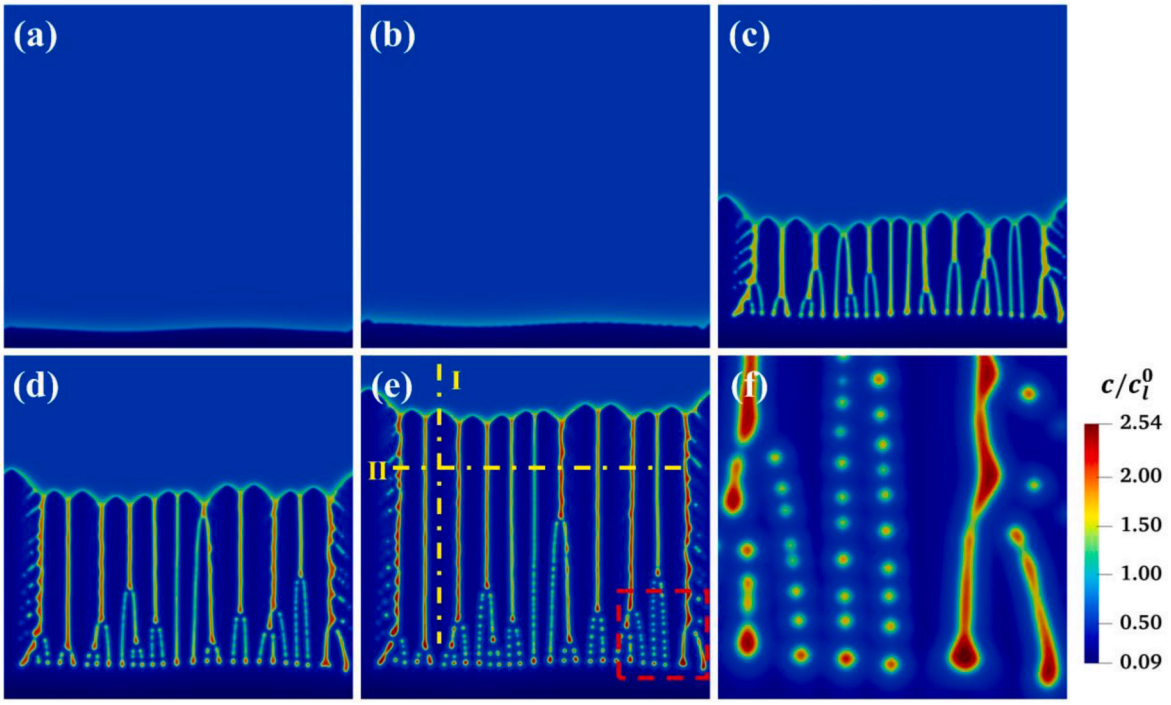



**Fig. 7.** Solute distribution at (a) $t = 8,000\Delta t$, (b) $t = 10,000\Delta t$, (c) $t = 20,000\Delta t$, (d) $t = 25,000\Delta t$, (e) $t = 30,000\Delta t$, (f) $t = 30,000\Delta t$ with scanning speed 4 mm/s.

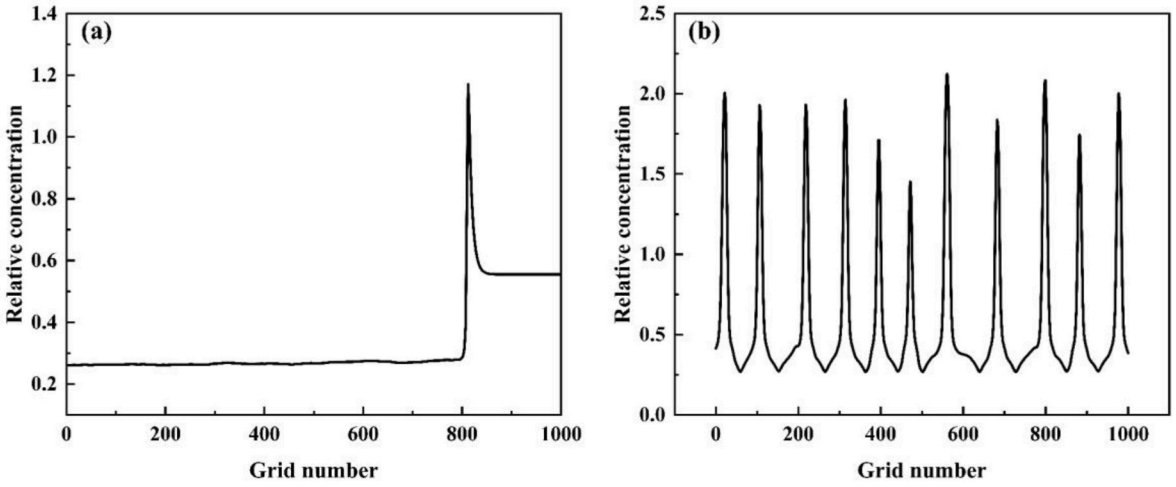



**Fig. 8.** Distribution of solute concentrations at dashed lines I and II in Fig. 7(e).

height and maximum temperature of the melt pool are inversely proportional to the scanning speed. That is, as the scanning speed increases, the height and maximum temperature of the melt pool decrease. Meanwhile, as shown in Fig. 5(d), we compared the melt pool morphology of the WAAM experimental results and the FE simulation results at different scanning speeds. With the increase in scanning speed, the errors between the experimental and finite element simulated melt pool depths are 0.257, 0.283, and 0.033 mm, respectively. It can be observed that the melt pool morphology matches well, thus verifying the accuracy of the finite element model.

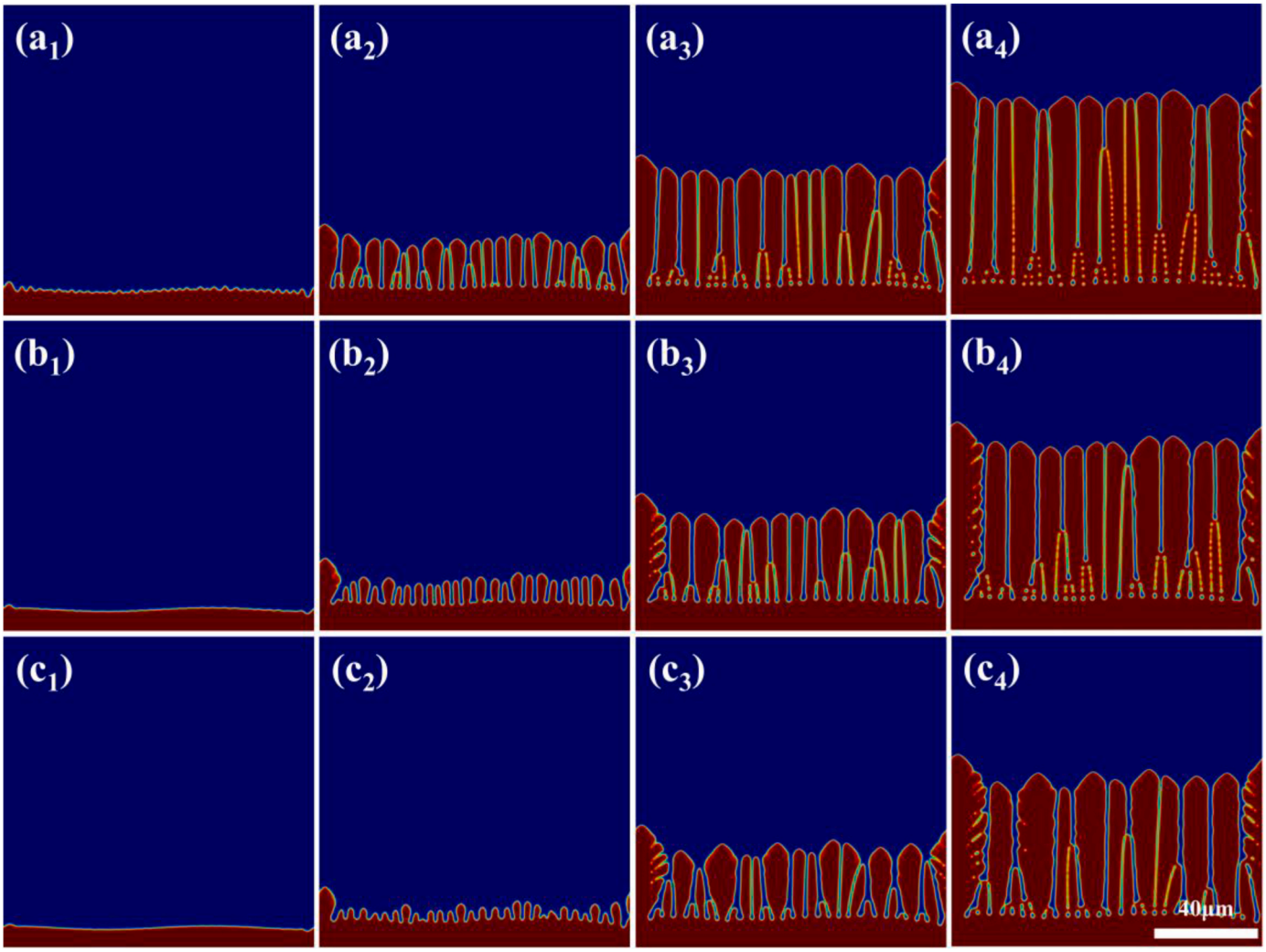



**Fig. 9.** Dendrite growth with scanning speed of ($a_1$) - ($a_4$) 3 mm/s, ($b_1$) - ($b_4$) 4 mm/s, ($c_1$) - ($c_4$) 5 mm/s at $t = \mathbf{10,000}\Delta t, \mathbf{15,000}\Delta t, \mathbf{20,000}\Delta t,$ **and** $\mathbf{25,000}\Delta t$.

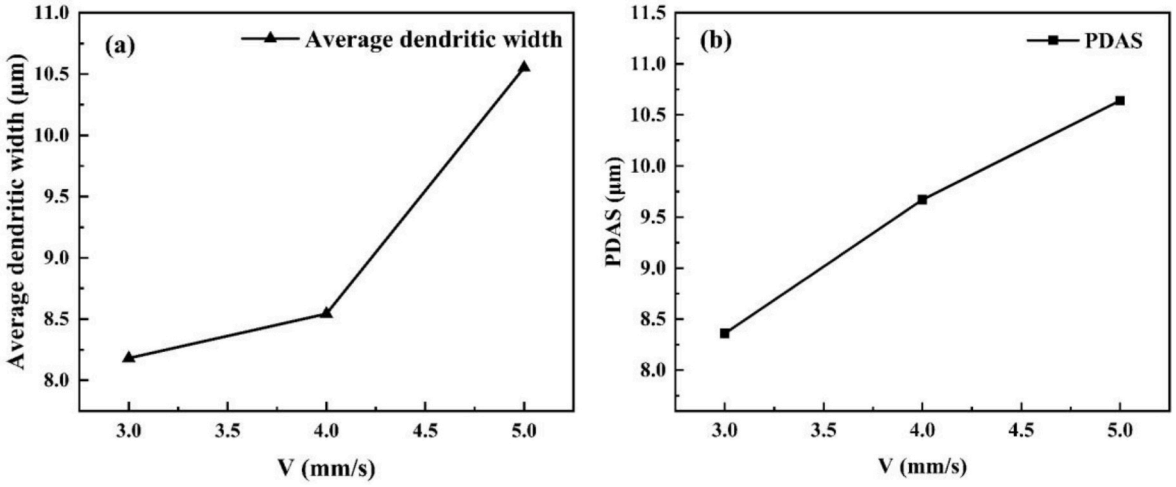



**Fig. 10.** Variation of average dendritic width and PDAS at different scanning speeds.

### 3.2. Phase field simulation of dendrite morphology and solute segregation

To verify the feasibility of the PF model [23] coupled with the fast solidification "antitrapping current", the dendrite growth and solute segregation at different scanning speeds are simulated in this section. The scanning speed during WAAM is an important parameter affecting the sample's microstructure. Fig. 6 shows the dendrite morphology simulated by PF when the current is 170 A and the scanning speed is 4 mm/s. A typical columnar dendrite structure appears during the WAAM process. In the solidification process, the solid-liquid interface first pushes toward the liquid phase in the form of a planar interface, as shown in Fig. 6 (a)–(b) [35,36]. As the solidification process proceeds, the solute atoms are continuously expelled into the liquid phase due to the diffusion coefficient of the liquid phase being larger than the solid phase, resulting in constitutional supercooling at the solid-liquid interface. Under the combined action of constitutional supercooling and temperature gradient, the solid-liquid interface is perturbed and produces protrusions, and the planar growth is transformed into columnar

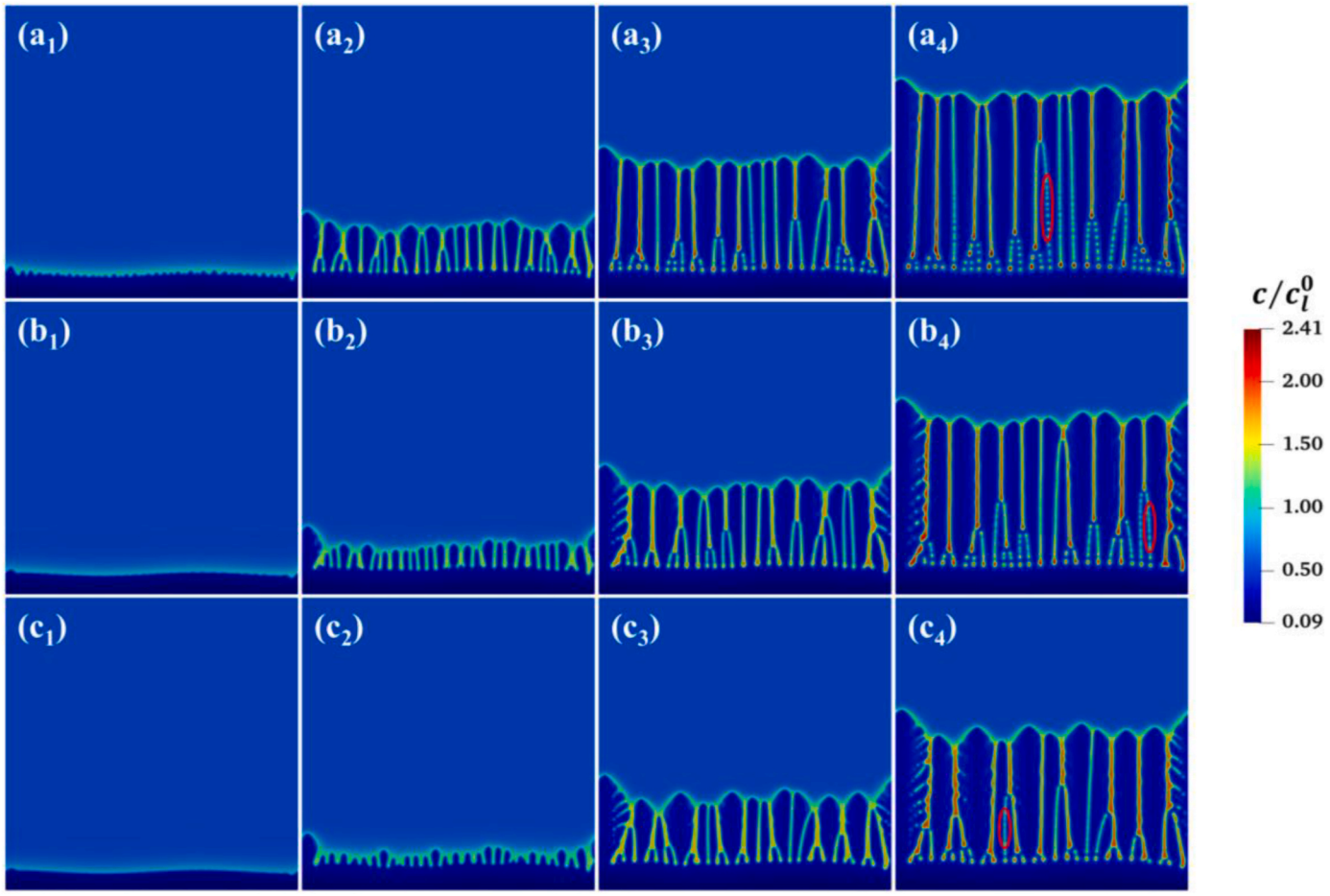



**Fig. 11.** Solute distribution with scanning speed of $(a_1)$ - $(a_4)$ 3 mm/s, $(b_1)$ - $(b_4)$ 4 mm/s, $(c_1)$ - $(c_4)$ 5 mm/s at $t = \mathbf{10,000}\Delta t, \mathbf{15,000}\Delta t, \mathbf{20,000}\Delta t,$ **and** $\mathbf{25,000}\Delta t$.

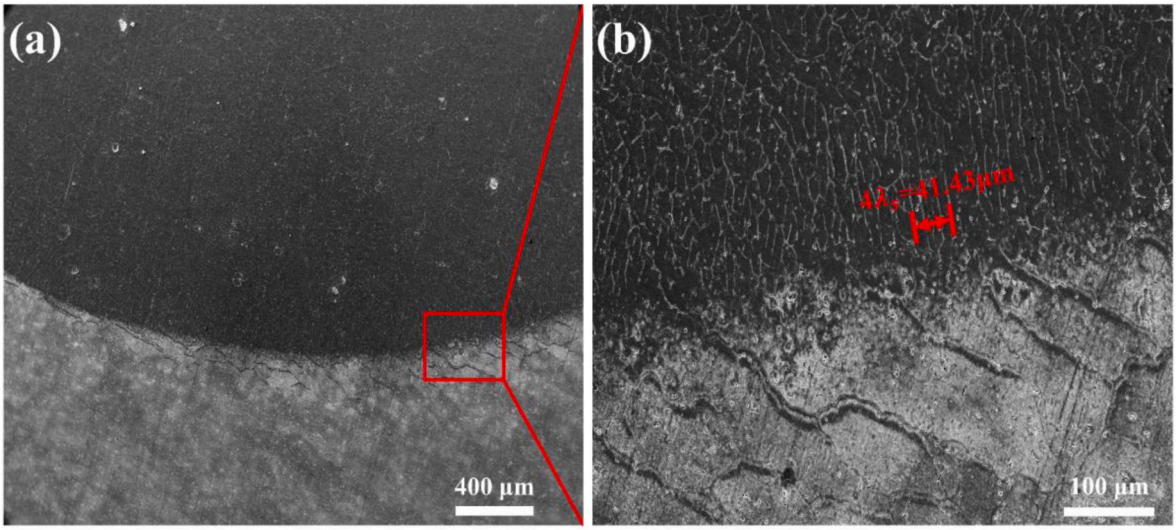



**Fig. 12.** SEM image of the specimen at a scanning speed of 4 mm/s.

dendrite growth, as shown in Fig. 6 (c)–(d). To reduce the interfacial energy, competitive growth occurs between primary dendrites, and some dendrites are fused, as shown in Fig. 6 (e). Dendrite growth tends to be steady-state growth after the end of the competitive growth phase, as shown in Fig. 6 (f).

Fig. 7 shows the solute distribution during the growth of columnar dendrites when the current is 170 A, and the scanning speed is 4 mm/s. During the transformation of the solid-liquid interface from planar to dendritic growth, the solute atoms are enriched at the front end of the solid-liquid interface and the interstices of the dendrites due to the difference in diffusion coefficients between the solid and liquid phases, as shown in Fig. 7 (a)–(d). In the solidification process, the segregation between the dendrites is aggravated, as shown in Fig. 7 (e)–(f). This is mainly because when the solute atoms are expelled into dendrite interstices, the solute atoms cannot diffuse sufficiently under the obstruction of dendrites, which makes the solute content between the dendrites higher than the dendrite tips. To quantitatively investigate the segregation behavior of the material during WAAM, the solute distribution at the yellow dashed lines (I and II) in Fig. 7 (e) was extracted. Fig. 8 (a) shows the solute distribution at the dashed line I. The solute content inside the dendrite in the path I remains almost constant due to the low solute diffusion rate in the solid phase. At the tip of the dendrite,

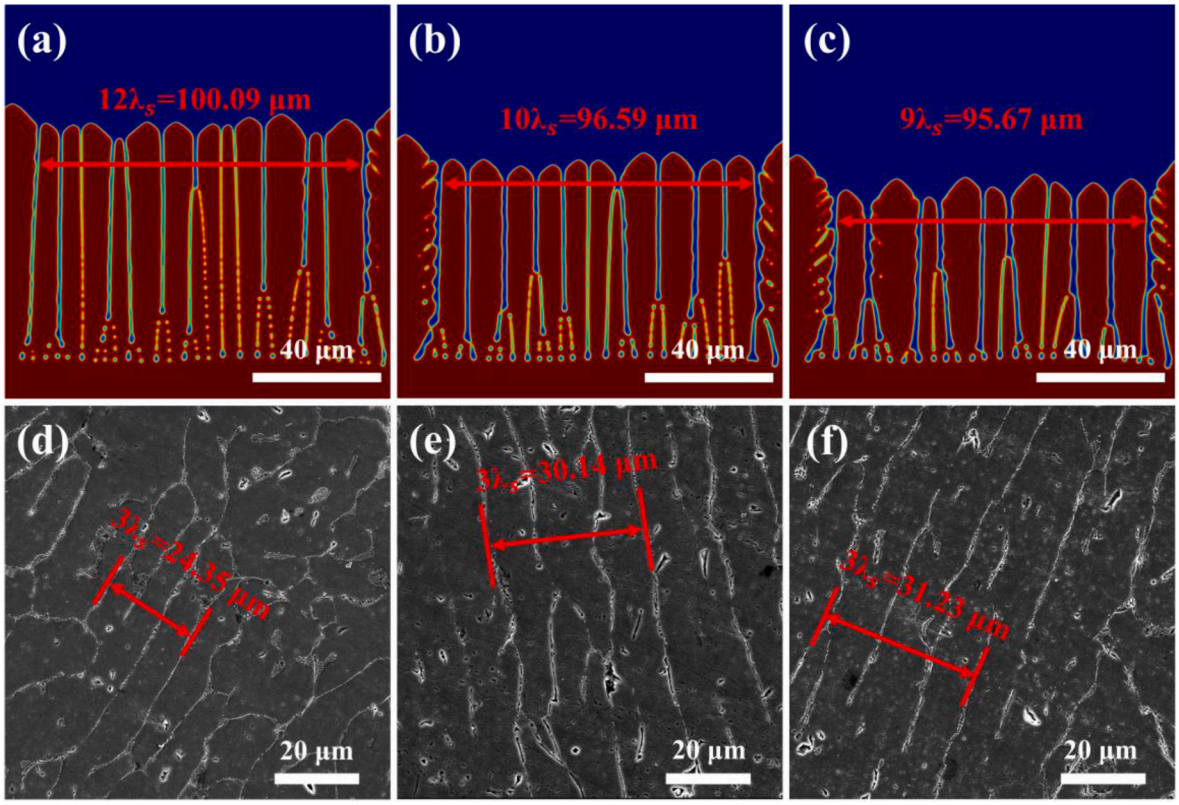



**Fig. 13.** (a)–(c) Simulation results of phase field at scanning speeds of 3 mm/s, 4 mm/s, and 5 mm/s, respectively, (d)–(e) SEM images at scanning speeds of 3 mm/s, 4 mm/s, and 5 mm/s, respectively.

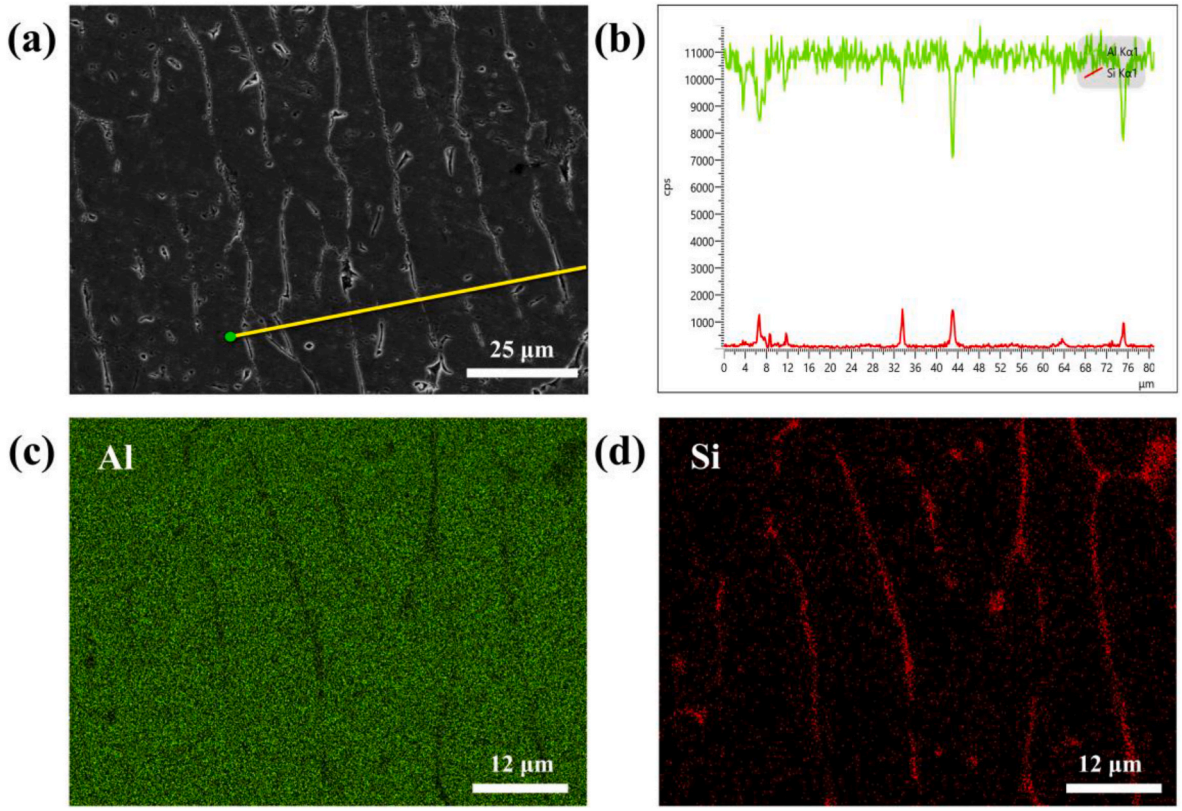



**Fig. 14.** EDS analysis at a scanning speed of 4 mm/s.

the solute atom concentration increases sharply at the solid-liquid interface and stabilizes in the liquid phase away from the solid-liquid interface. Fig. 8 (b) depicts the solute distribution at the dashed line II. It can be observed that the solute content distribution shows a typical "U" -shaped structure in the transition from the dendrites to the dendritic interstices, and the content of solute atoms increases sharply at the dendritic interstices.

Fig. 9 shows the comparison of PF simulation results at different scanning speeds when the current is 170 A. During WAAM, the scanning speed significantly affects the temperature gradient during solidification. As the scanning speed increases, the temperature gradient during dendrite growth decreases, and the driving force for dendrite growth decreases, so it can be observed that the dendrite growth velocity decreases as the scanning speed increases. To reflect more intuitively the dendrite growth at different scanning speeds, the primary dendrite arm spacing (PDAS) $\lambda_s$ is used to describe the dendrite growth, $\lambda_s = L/n$, where $L$ is the measurement distance and $n$ is the number of dendrites within the measurement distance. Fig. 10 shows the PDAS and average dendrite width for different scanning speeds. The scanning speed is proportional to the PDAS and the average dendrite width, which means PDAS and average dendrite width increase as the scanning speed increases. The increase of the scanning speed during WAAM leads to a decrease in the temperature gradient, and the driving force of the dendrite growth along the temperature gradient direction is weakened, resulting in the growth of the dendrite in the direction perpendicular to the temperature gradient.

Fig. 11 shows the solute distribution at different scanning speeds, and it can be observed that the segregation between dendrites decreases with increasing scanning speed. Decreasing the temperature gradient allows sufficient time for the solute atoms to diffuse. As shown in the red boxes in Fig. 11 ($a_4$), ($b_4$), and ($c_4$), discontinuous droplets are formed during solidification. As solidification proceeds, dendrite coarsening adjacent dendrites fuse, forming a solid lattice [37]. Since columnar dendrites form in the fusion line region between the deposited layer and the substrate during WAAM, the formation of this solid lattice makes liquid feeding difficult. This leads to an increased susceptibility to solidification cracking, which causes cracks in the fusion line region.

### 3.3. Experimental results

Fig. 12 shows the SEM image of the specimen at a current of 170 A and a scanning speed of 4 mm/s. It can be seen that near the fusion line, the dendrites show columnar dendrite growth along the direction of the temperature gradient. The PDAS is 10.36 μm, as shown in Fig. 12 (b), which is in good agreement with the simulated result of 9.66 μm. The growth morphology of columnar dendrites simulated by PF at different scanning speeds with SEM observations are shown in Fig. 13. From the PF simulation results in Fig. 13 (a)–(c), it can be observed that the PDAS increases with the increase of scanning speed. The same conclusion can be obtained from the SEM observations in Fig. 13 (d)–(f), where the characteristics of the columnar dendrite morphology from the PF simulations are in good agreement with the experimental observations. It illustrates the feasibility of the improved PF model applicable to additive manufacturing in predicting the effect of processing parameters on dendrite growth.

Fig. 14 shows the EDS analysis of the elemental distribution of the specimen at a current of 170 A and a scanning speed of 4 mm/s. The Si elements are mainly distributed at the grain boundaries, while the content of Al elements at the grain boundaries is low, which is consistent with the simulation results (as shown in Fig. 7).

## 4. Discussion

### 4.1. Evolution of dendrite morphology at different scanning speeds

A large temperature gradient (usually $>10^4$ K/m [21]) exists during WAAM. With the solidification process, solute atoms are continuously expelled into the liquid, resulting in the enrichment of solute atoms at

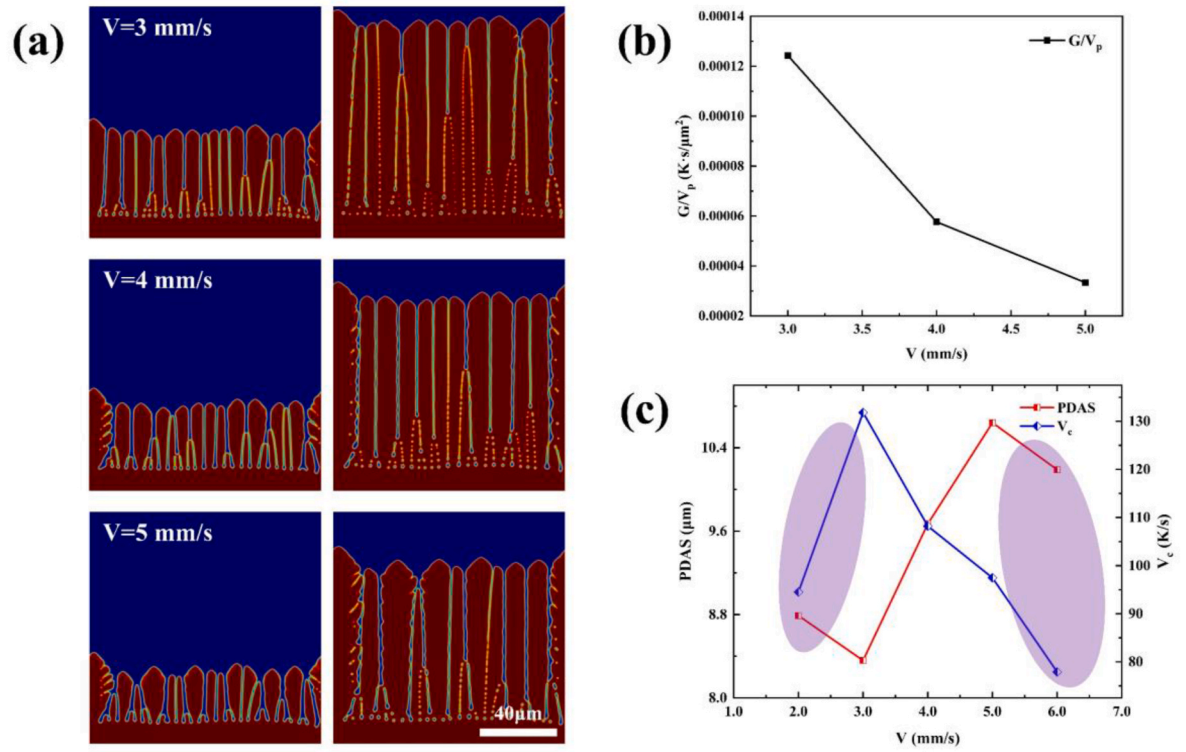



**Fig. 15.** Effect of $G(t)$ and $V_p$ on dendrite growth at different scanning speeds.

the solid-liquid interface. At the same time, the solid-liquid interface is disturbed by a large temperature gradient, which causes the solid-liquid interface to produce protrusions and grow in the direction of the temperature gradient. Since the temperature gradient along the vertical direction is much larger than the horizontal direction, the number of secondary dendrites produced during the growth of columnar dendrites is smaller. The enrichment of solute atoms in the interstices also inhibits the growth of secondary dendrites. After the dendrite enters the

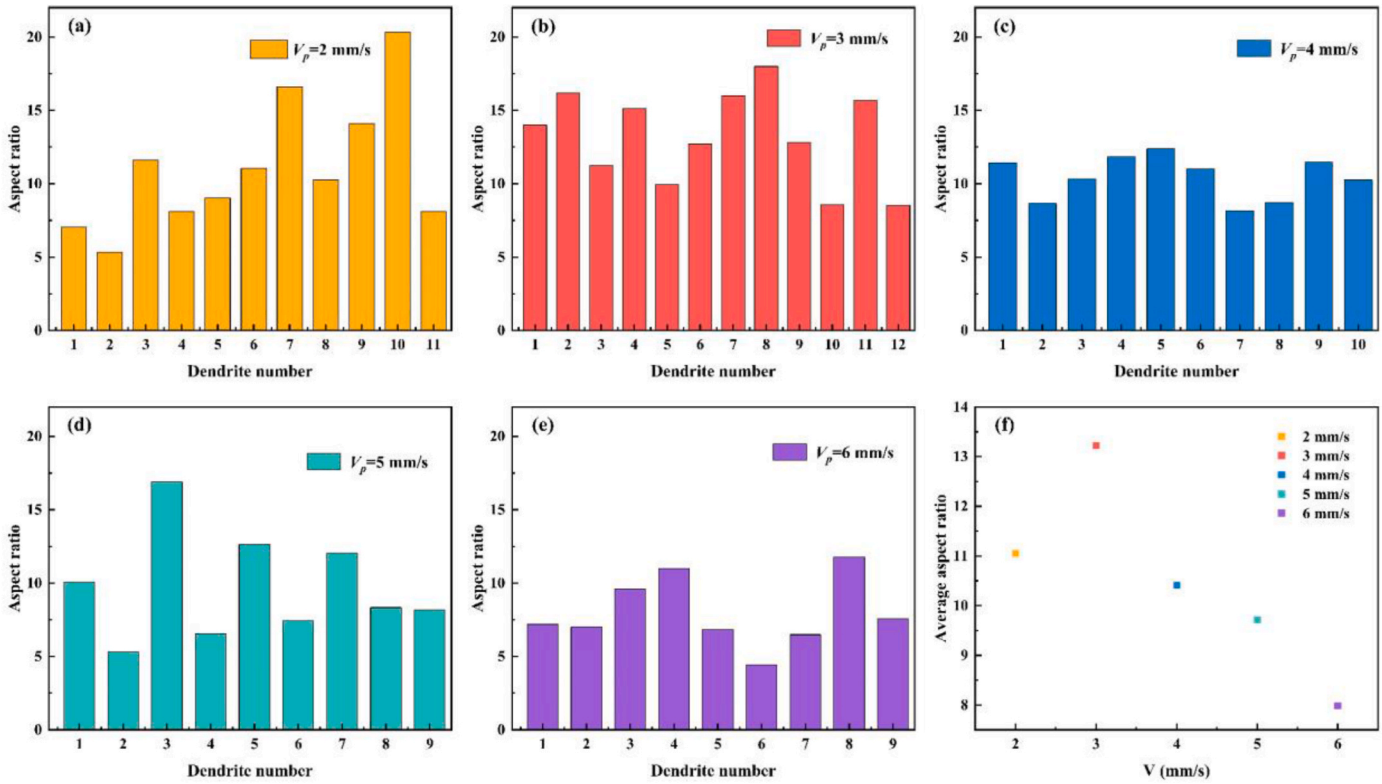



**Fig. 16.** Aspect ratio of dendrites at different scanning speeds.

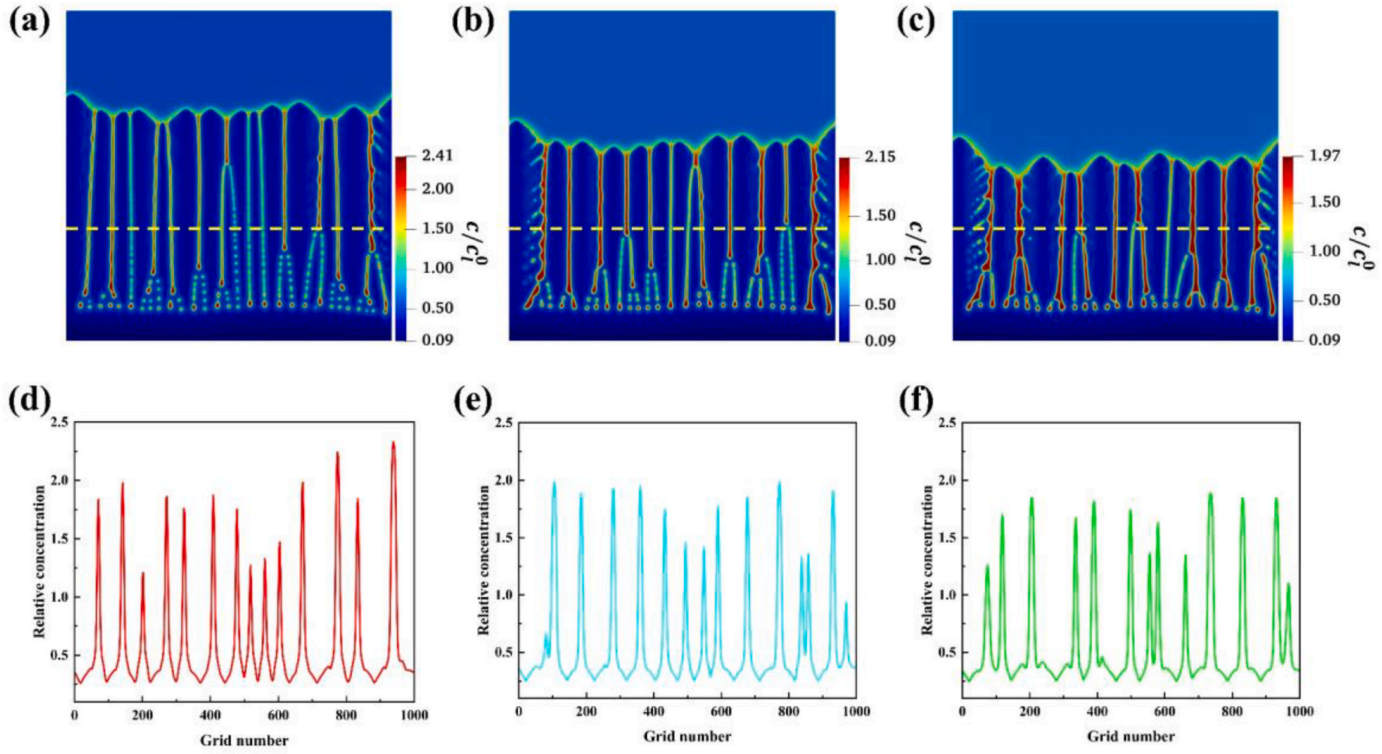



**Fig. 17.** Effect of scanning speed on solute distribution between dendritic interstices.

steady-state growth stage, the protrusion of secondary dendrites is inhibited by the combined effect of temperature gradient and solute segregation, as shown in Fig. 9. Zhang et al. [38] suggested that it may be due to the small spacing of primary dendrites that hinders the growth of lateral branches or the large $G(t)/V_p$. Fig. 15(a) shows the dendrite growth at different scanning speeds. As shown in Fig. 15 (b), it can be found that $G(t)/V_p$ gradually decreases as the scanning speed increases, and the secondary dendrite development gradually increases as $G(t)/V_p$ decreases, as shown in Fig. 15 (a). This is consistent with the theory of Zhang et al. [38].

The change in scanning speed during WAAM affects both the temperature gradient $G(t)$ and solidification velocity $V_p$, as shown in Eqs. (7) and (8). Farzadi et al. [39] demonstrated the effect of cooling rate ($V_c = G(t)V_p$) on the growth of columnar dendrites, and it was found that the PDAS gradually decreased with increasing cooling rate. Fig. 15 (c) shows the cooling rate $V_c$ and PDAS at different scanning speeds. A higher scanning speed produces a higher solidification speed, which leads to increased movement of the solid-liquid interface and faster growth of columnar dendrites. However, the simulation results show that the growth velocity of columnar dendrites slows down as the scanning speed increases, as shown in Fig. 15 (a). This is because the increase in scanning speed also leads to a decrease in the temperature gradient, which makes the driving force of dendrite growth decrease and the growth velocity of columnar dendrite decrease. The columnar dendrites grow under the combined effect of temperature gradient and solidification speed. Within a certain range, the growth of columnar dendrites is more sensitive to temperature gradients.

However, PDAS does not increase or decrease without limit during WAAM. We compare the PDAS and $V_c$ when $V$ is 2 mm/s and 6 mm/s, as shown in Fig. 15 (c). It can be found that a lower $V$ (2 mm/s) leads to a lower $V_c$. Although $G(t)$ is higher in the WAAM process at a lower $V$, a lower $V_p$ limits the growth of dendrites and thus leads to a higher PDAS. As expected, a higher $V$ (6 mm/s) produces a lower $V_c$ , yet the change in the PDAS is different from our previous conclusions. The WAAM process, at relatively high $V$, $V_p$ replaces the dominance of $G(t)$ in influencing the growth of dendrites, leading to a decrease in PDAS.

The aspect ratios of the columnar dendrites at different scanning speeds of $30,000\Delta t$ are shown in Fig. 16. The trends of the dendrite aspect ratio and cooling rate are consistent. The low solidification speed, when $V$ is 2 mm/s, inhibits the increase of cooling rate, resulting in a decrease in dendrites with high aspect ratios, as shown in Fig. 16(a). Meanwhile, when $V = 3$ mm/s - 6 mm/s, the columnar grains within the scanning track are refined with the increase in scanning speed, and the columnar dendrites decrease in the aspect ratio, as shown in Fig. 16 (b)–(e). The fine grains will improve the mechanical properties of WAAM products. A similar trend of the effect of increasing scanning speed on the strength and plasticity of Al–Cu alloys was observed in the experimental work of Geng et al. [16]. This indicates that the reduced cooling rate mitigates solute segregation, resulting in higher mechanical properties.

### 4.2. *Solute segregation at different scanning speeds*

During rapid solidification, the higher temperature gradient leads to insufficient time for solute atoms to diffuse, thus allowing solute atoms to be enriched at the dendrite interstices. Fig. 17 (d), (e), and (f) show the solute distribution extracted from the same height of Fig. 17 (a), (b), and (c) along the dashed positions. It can be found that the density of U-shaped peaks decreases with increasing scanning speed. This is mainly due to the average dendrite width and PDAS increase with increasing scanning speed, resulting in a decrease in the number of dendrites in the same simulation domain. The highest values of the relative concentration of Si between dendrites are 2.33, 1.99, and 1.89 when the scanning speed is 3 mm/s, 4 mm/s, and 5 mm/s, respectively, as shown in Fig. 17 (d), (e), and (f). An increase in the scanning speed leads to a decrease in the growth velocity of columnar dendrites, and the decrease in the growth velocity of dendrites facilitates the diffusion of solute atoms compared to the growth of dendrites at high growth velocity. Therefore, the segregation effect of solute atoms decreases as the scanning speed

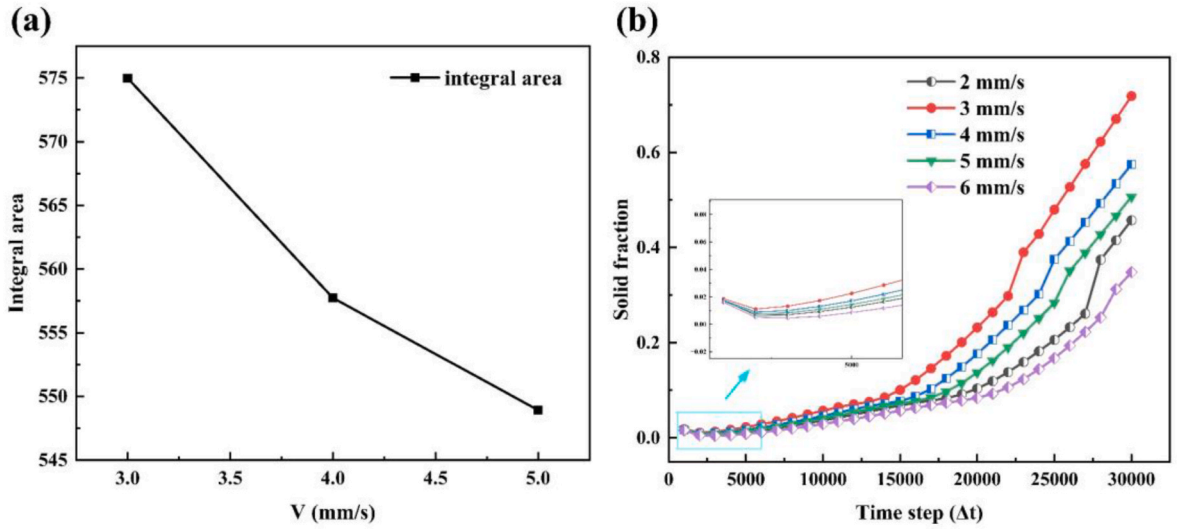



**Fig. 18.** (a) Fig. 17(d), (e), (f) integral areas of solute distribution curves, (b) Solid fraction in dendrite growth.

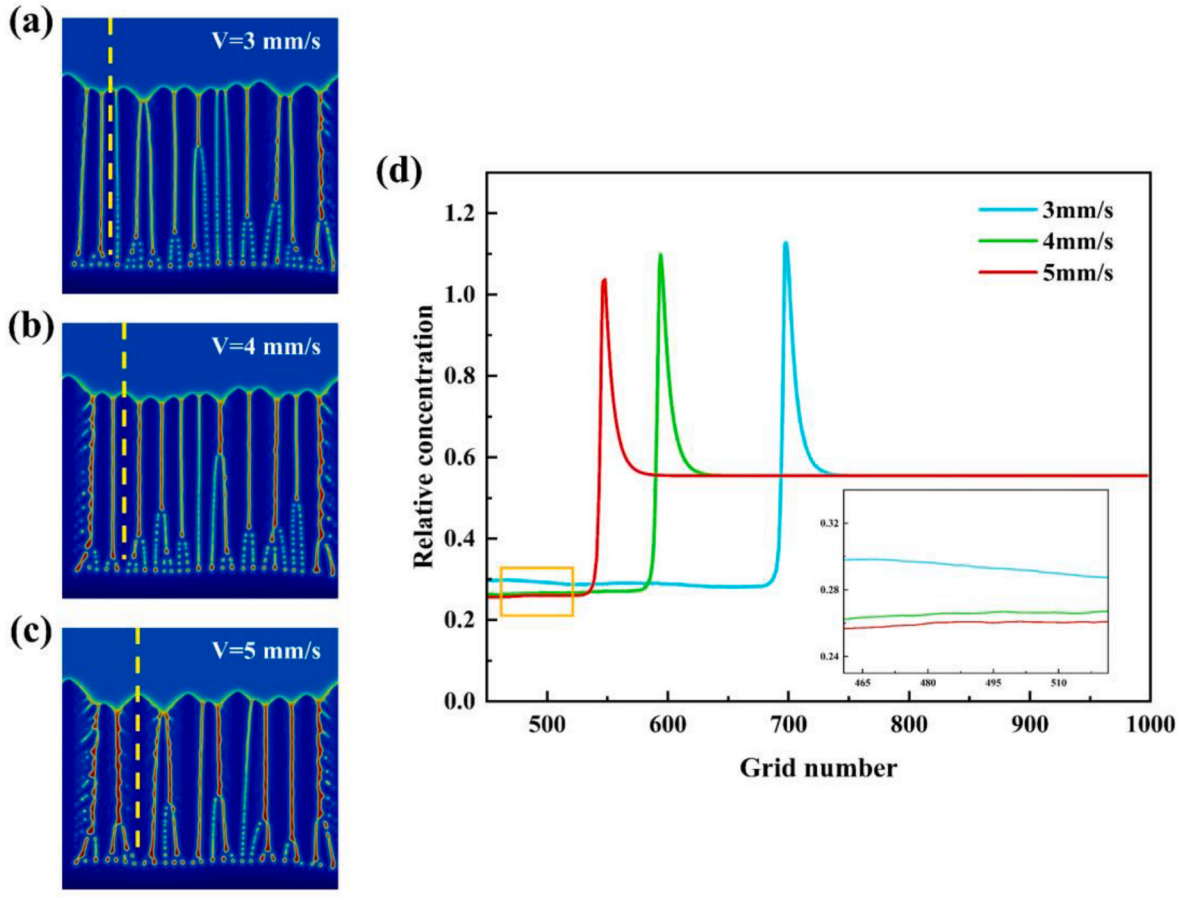



**Fig. 19.** Effect of scanning speed on the solute distribution at the tip of dendrites.

increases. The solute segregation is related to the partition coefficient, and the solute partitioning under slow solidification is close to the equilibrium phase diagram in the quantitative PF model proposed by Karma et al. [20]. However, during WAAM, solidification is fast, and the diffusion of solute atoms at the solid-liquid interface cannot reach local equilibrium, which leads to the formation of supersaturated phases [31], resulting in a non-physically high concentration of solute atoms at the solid-liquid interface.

Fig. 18(a) shows the integral area of the solute distribution curves of Fig. 17(d), (e), and (f). As the scanning speed increases, the solute segregation within the scanning trajectory decreases, and the temperature gradient decreases, allowing for adequate solute diffusion. The cooling rate (Fig. 15(c)) is the primary factor affecting dendrite growth, as seen from the change in solid phase fraction in dendrite growth, as shown in Fig. 18(b). A high dendrite growth rate implies a shorter solidification time and a more inhomogeneous solute distribution. At the

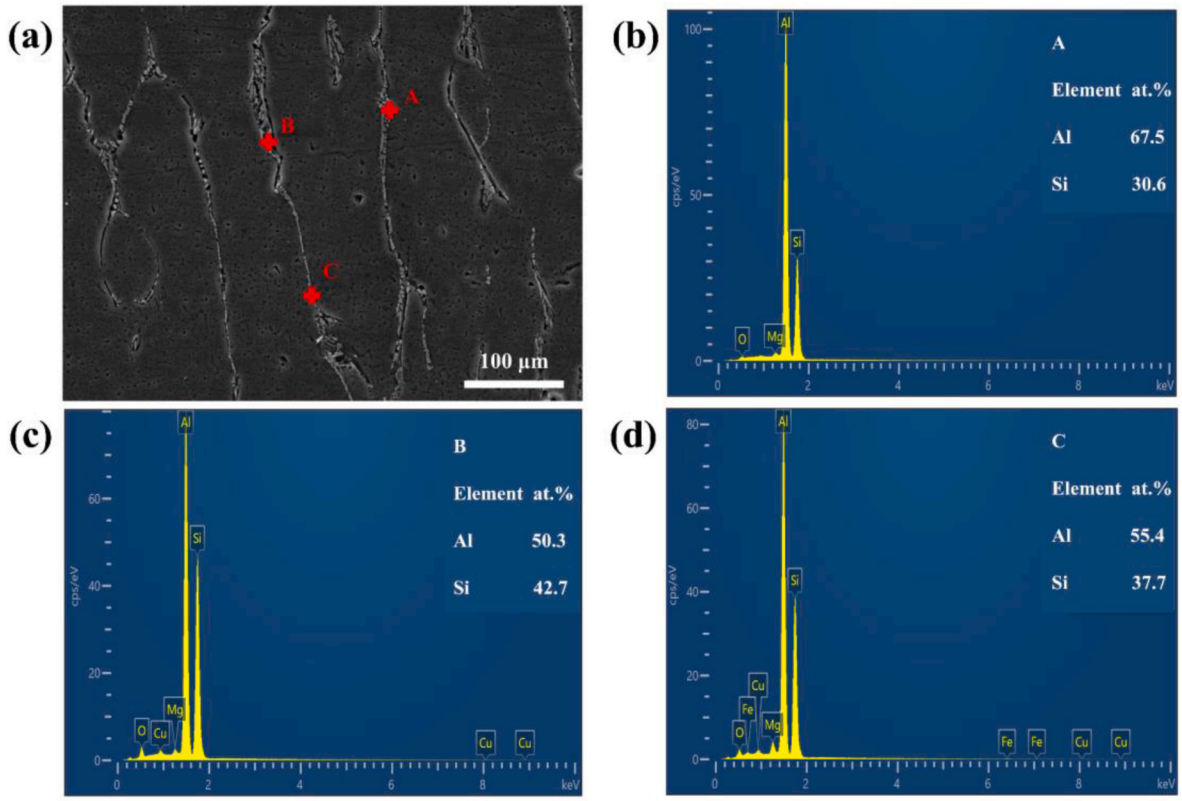



**Fig. 20.** EDS-point results of WAAM specimens at a scanning speed of 3 mm/s.

same time, the high aspect ratio columnar dendrites have stronger anisotropy in the direction of dendrite growth versus other directions, which affects the quality of WAAM parts. Fig. 19 (d) shows the solute distribution extracted along the dashed lines in Fig. 19 (a), (b), and (c). The solute field can be divided into a solid concentration area, a solute enrichment area, and a liquid concentration area [40,41]. As shown in Fig. 19 (d), it can be found that the solute concentration at the dendrite tip decreases with the increase of scanning speed, which is consistent with the conclusion reached previously in this paper. Meanwhile, the solute atom content within the dendrite decreases with increasing scanning speed, as shown in the yellow box in Fig. 19 (d). As the scanning speed increases, the temperature gradient decreases, which causes some of the solute atoms to not diffuse sufficiently during solidification and to be "trapped" in the solid phase.

To compare the simulated solute segregation with the degree of solute segregation in the actual WAAM process, Fig. 20 shows the EDS-point results when the scanning speed is 3 mm/s. The three points A, B, and C in Fig. 20 (a) were selected to quantify the solute content at the grain boundaries, and the obtained elemental analysis results are shown in Fig. 20 (b), (c) and (d). It can be observed that the elemental content of Si at the grain boundaries of A, B, and C is 30.6 at%, 42.7 at%, and 37.7 at%, respectively. As shown in Fig. 17 (d), the relative values ($c/c_l^0$) of the solute content between the dendrites simulated by the PF when the scanning speed is 3 mm/s range from 1.21 to 2.33, meaning the solute content of elemental Si between the dendrites is 30.99 at% to 69.01 at%. The experimental and simulation results obtained an excellent agreement.

## 5. Conclusion

In this work, combined with a 3D FE model, using a quantitative PF model improved by Kavousi et al. [23], applicable to AM conditions, we performed PF simulations of dendrite growth and solute segregation in the WAAM process with different scanning speeds under constant current conditions. Additionally, experimental verification was conducted to ensure the reliability of our simulation results. The main conclusions are as follows.

(1) The change of scanning speed $V$ during WAAM affects both the solidification speed $V_p$ and temperature gradient $G(t)$. Among them, the solidification speed $V_p$ increases, and the temperature gradient $G(t)$ decreases as the scanning speed increases. At the same time, the dendrite growth velocity decreases with the increase in scanning speed. It indicates that the temperature gradient plays a dominant role in the growth of columnar dendrites within a certain range.
(2) An increase in the scanning speed $V$ leads to a decrease in $G(t)/V_p$, which makes the secondary dendrites more developed. Meanwhile, higher scanning speeds within a specific range lead to a lower temperature gradient, which produces higher PDAS and average dendrite width as well as lower aspect ratios.
(3) The lower scanning speed $V$ will increase the temperature gradient and the driving force of dendrite growth, resulting in a weaker diffusion of solute atoms and an increase in solute enrichment. This may lead to the formation of discontinuous solid-phase regions and the increase of crack sensitivity.
(4) By comparing with the experimental results, the improved PF model can accurately predict the dendrite growth and solute segregation in the WAAM process, establishing a quantitative link between the macroscopic process and microstructure and laying a foundation for the rational formulation of the WAAM process.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgements

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